

Learning Notes

Math Foundations of Agent-X

*A ground-up guide to the proof graph,
for someone with no math background.*

Generated from pathram2 composable sections
doc-14 (learning-notes) with 13 linked sections

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Part 1: Sets

A SET is a collection of distinct things.

Rules

Rule	Example	Explanation
No duplicates	$\{1, 1, 2\} = \{1, 2\}$	Repeats collapse
Order irrelevant	$\{3, 1, 2\} = \{1, 2, 3\}$	Same elements = same set
In or not in	$3 \in \{1, 2, 3\}, 5 \notin \{1, 2, 3\}$	No "half in"

Operations on Sets

Operation	Symbol	Example	Result
Union	$A \cup B$	$\{1, 2\} \cup \{2, 3\}$	$\{1, 2, 3\}$
Intersection	$A \cap B$	$\{1, 2\} \cap \{2, 3\}$	$\{2\}$
Difference	$A \setminus B$	$\{1, 2, 3\} \setminus \{2\}$	$\{1, 3\}$
Product	$A \times B$	$\{1, 2\} \times \{x, y\}$	4 pairs

Key Concepts

Concept	Meaning	Example
Subset	Every element of A is in B	$\{1, 2\} \subseteq \{1, 2, 3\}$
Empty set	Nothing; subset of everything	$\emptyset \subseteq \text{anything}$
Cardinality	How many elements	3 elements

Part 2: Relations

A RELATION is a rule applied to two elements yielding yes or no. Properties describe the RULE, not the set.

Three Key Properties

Property	Question	=	<	$\leq$	"is parent of"
Reflexive	Every element relates to itself?	Yes (5=5)	No (5<5?)	Yes (5 $\leq$ 5)	No
Symmetric	If $A \rightarrow B$, does $B \rightarrow A$?	Yes	No (3<5 but 5 $\not<$ 3)	No	No
Transitive	If $A \rightarrow B$ and $B \rightarrow C$, does $A \rightarrow C$?	Yes	Yes	Yes	No

Combined Structures

Structure	Properties	Example	In Agent-X
Equivalence relation	reflexive + symmetric + transitive	"=" (sameness)	abheda (non-difference)
Partial order	reflexive + antisymmetric + transitive	" $\subseteq$ " (subset)	sthita (rests-on)

Part 3: Operations

An OPERATION takes inputs and produces an output.

Type	Inputs	Example
Unary	one	neg(5)=-5, square(3)=9
Binary	two	add(3,5)=8, mul(4,3)=12

Properties of a Binary Operation

Property	Rule	+ example	- example
Closure	Result stays in set	int+int=int ✓	nat-nat=maybe negative ×
Associativity	$(a \circ b) \circ c = a \circ (b \circ c)$	$(2+3)+4 = 2+(3+4)$ ✓	$(5-3)-1 \neq 5-(3-1)$ ×
Identity	Element that does nothing	$0 (x+0=x)$	—
Inverse	Element that undoes	-5 undoes 5	—
Commutativity	$a \circ b = b \circ a$	$3+5=5+3$ ✓	$3-5 \neq 5-3$ ×

Part 4: Algebraic Structures

Set + operation + properties = structure. Each level adds one requirement.

Hierarchy

Set + closure + associativity = Semigroup
 + identity = Monoid
 + inverses = Group
 + commutativity = Abelian Group
 Two ops: $\oplus$ =abelian group, $\otimes$ =monoid, distributive = Ring
 + layers = Graded Ring

Monoid (closure + associativity + identity)

Set	Operation	Identity	Inverses?
naturals	+	0	No (-3 not in set)
naturals	x	1	No (1/5 not in set)
All strings	concatenation	""	No
All lists	append	[]	No

Group (monoid + inverses)

Set	Operation	Identity	Inverse of 5
integers	+	0	-5
rational (nonzero)	x	1	1/5
rotations 0/90/180/270	rotation	0 deg	270 deg

Ring (two interacting operations)

$\oplus$ (first)	$\otimes$ (second)
Structure	Abelian group
Identity	0
Inverses	yes (required)
Commutative	yes (required)
Interaction	$\otimes$ distributes: $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$

In Agent-X:

Ring element	Math	Sanskrit	Meaning
$\oplus$	addition	yukta	connection/gathering
$\otimes$	multiplication	kriya	action/transformation
0	zero	shunya	emptiness
1	one	swarupa	self-nature
-1	negation	pratipaksha	opposition

Graded Ring (ring with layers)

$R = R_0 \oplus R_1 \oplus R_2 \oplus \dots$ Each layer is a grade.

Polynomial example:

Polynomial	What is in the grade
Grade 0	constants: 5, -3
Grade 1	linear: $2x$, $-x$
Grade 2	quadratic: $3x^2$
$\oplus$ within grade	$3x^2 + 5x^2 = 8x^2$
$\otimes$ across grades	$2x \times 3x^2 = 6x^3$

Agent-X paragraph — each sentence is a grade, but a grade is NOT just "facts". Each grade is a rich set of typed triples across multiple floors:

Example: "ball-A has mass 5 kg. 3 birds flew away. find kinetic energy."

Grade R_0 (sentence 1)	Grade R_1 (sentence 2)	Grade R_2 (sentence 3)
Entity	ball-A (naama-mudra)	bird (naama-mudra)
Number	5 (sankhya)	3 (sankhya)
Unit	kg (matra)	—
Binding	mass to 5kg (rashi-bandha)	—
Signal	—	kshaya/decrease
Subject	ball-A (prathama)	bird (prathama)
Tense	present	past

Each grade contains 1s across MANY floors simultaneously. A grade is not one floor — it is a horizontal slice of the tensor at a particular sentence, spanning all 62 floors.

Operation	What it does	Example
$\oplus$ within a grade	OR (accumulate)	"mass AND velocity" in same sentence
$\otimes$ across grades	chain/select	"the second joint" references entity from earlier grade
Grade boundary	viraam (period)	resets intra-sentence indexing
Distributivity	"respectively"	"joints 1 and 2 have masses 5 and 10 respectively"
Identity (0)	shunya	empty accumulator before first sentence

Part 5: Morphisms and Composition

Morphism (structure-preserving map)

Rule: $f(a \oplus b) = f(a) \oplus f(b)$. Combine then map = map then combine.

Function	Preserves?	Test	Morphism?
$f(x)=2x$	addition	$f(3+5)=16$, $f(3)+f(5)=16$	Yes
$f(x) = x^2$	addition	$f(3+5)=64$, $f(3)+f(5)=34$	No
$\text{length}(s)$	concat to add	$\text{len}("ab"+"cd")=4=2+2$	Yes
$\log(x)$	mul to add	$\log(ab)=\log(a)+\log(b)$	Yes

Types: homomorphism (one-way), isomorphism (perfect two-way), endomorphism (to itself).

In Agent-X the lexical morphism δ maps English to graph concepts:

Word	δ maps to	Meaning
"heavier"	viveka-max	discrimination-maximum
"flew away"	kshaya	decrease
"find"	vidhi-kaala	seek/imperative
"more"	vridhhi	increase

δ preserves structure: "more" maps to vridhhi, "flew away" maps to kshaya, and $G[vridhhi][kshaya][pratipaksha]=1$.

Composition (chaining)

$(g \circ f)(x) = g(f(x))$. Do f first, then g. Associative but NOT commutative.

In Agent-X a krama IS a composition:

Formula	Krama	Steps
$K = \frac{1}{2}mv^2$	half, mul, square	square(v), mul(m,..), half(..)
$p=mv$	mul	mul(m,v)
Solve K for v	sqrt, div, double	double(K), div(..,m), sqrt(..)

Inversion = reverse composition + pratipaksha each step.

Part 6: The 3D Tensor

$G \in \{0, 1\}^{N \times N \times R}$ — a 3D box of 0s and 1s.

Axis	Size	Meaning
Rows	$N = 1604$	FROM which node
Columns	$N = 1604$	TO which node
Depth	$R = 62$	WHICH relation type (floor)

Visualize as 62 stacked grids, each 1604×1604 :

Floor 62
...
Floor 2
Floor 1

$$\begin{array}{c} 1604 \\ \times \\ 1604 \end{array}$$

Edges and Nodes

An EDGE = a 1 in the tensor: $G[\text{addition}][\text{subtraction}][\text{pratipaksha}] = 1$ — this IS the edge. $G[\text{addition}][\text{multiplication}] = 0$ — no edge.

A NODE = two vertical slices through all 62 floors: Outgoing: $G[\text{node}][*][*]$ — what this node points TO. Incoming: $G[*][\text{node}][*]$ — what points AT this node. Full identity = both slices combined.

Sparsity

About 37,000 ones in about 159 million cells = 0.023 percent filled. CSR (Compressed Sparse Row) stores only the 1s. Adding a node: $O(\text{edges on that node})$, typically 5-10. Microseconds. Density decreases as nodes grow (edges grow as N , cells grow as N^2).

Boolean Ring

Elements are 0 and 1. $\oplus$ = OR, $\otimes$ = AND: $0 \text{ OR } 0 = 0$, $0 \text{ OR } 1 = 1$, $1 \text{ OR } 1 = 1$ (accumulate). $0 \text{ AND } 0 = 0$, $0 \text{ AND } 1 = 0$, $1 \text{ AND } 1 = 1$ (select). $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ (distributivity). Identity for OR = 0 (shunya). Identity for AND = 1 (swarupa).

Part 7a: The 10 Static Generators

The visheshanam ring has 10 generators. Every dynamic dimension is composed from these.

#	Name	English	Floor question	Algebraic role	Example
1	swarupa	self-nature / IS	What IS this?	$1 \otimes$ (mul identity)	derivative IS jivamsha
2	abheda	non-difference / EQUALS	Same as what?	$\equiv$ (equivalence)	addition EQUALS sankalana
3	drishthanta	example / WITNESS	Concrete instance?	$\exists$ (existential)	monoid EXEMPLIFIED BY addition
4	sthita	foundation / RESTS ON	Built on what?	$\leq$ (partial order)	ring RESTS ON group
5	yukta	connection / CONNECTED TO	Associated with?	$\oplus$ (addition)	ring CONNECTED TO monoid
6	siddha	established / PROVEN BY	What verifies?	$\vdash$ (provability)	ring PROVEN BY distributivity
7	kriya	action / ACTS THROUGH	What mechanism?	$\otimes$ (multiplication)	derivative ACTS THROUGH change
8	phala	fruit / PRODUCES	What result?	$\rightarrow$ (consequence)	derivative PRODUCES rate, slope
9	janya	born-from / ORIGINATES	What inputs?	$\leftarrow$ (origin)	ideal BORN FROM ring
10	pratipaksha	opposite / INVERTS	What inverse?	$^{-1}$ (group inverse)	addition INVERTS subtraction

Ring structure

Property	$\oplus$ = yukta	$\otimes$ = kriya
Closure	✓	✓
Associativity	✓	✓
Identity	shunya (0)	swarupa (1)
Inverses	rahita (absence)	not required
Commutative	✓ (A connected to B = B connected to A)	$\times$ ("A acts through B" $\neq$ "B acts through A")

Internal pratipaksha pairs among generators

Generator	Pratipaksha	Meaning
phala (produces)	janya (born from)	output $\leftrightarrow$ input
yukta (connection)	rahita (absence)	presence $\leftrightarrow$ absence
kriya (action)	nirodha (cessation)	doing $\leftrightarrow$ stopping
abheda (sameness)	vibheda (distinction)	unity $\leftrightarrow$ difference

Part 7b: The 52 Dynamic Dimensions

Generated by composing the 10 static generators. Grouped by function.

Grammar / Morphology

#	Name	English	Captures	Example
11	sandhi	joining	morpheme combination	"running" = run+ing
12	dhatu	verb root	stripped inflection	"flew" $\rightarrow$ fly
13	vachana	number	singular/plural	"birds" = plural
14	purusa	person	1st/2nd/3rd	"I" vs "he"
15	naama-mudra	name stamp	entity recognition	"ball-A"
16	naama-pratibodha	name awakening	resolved entity	ball-A $\rightarrow$ object
17	dvandva	pair compound	two concepts joined	"space-time"

Vibhakti (Case — WHO did WHAT to WHOM)

#	Name	Case	Question	Example
18	prathama	nominative	WHO?	"THE CAR moves"
19	dvitiya	accusative	WHAT?	"Find THE ENERGY"
20	trtiya	instrumental	BY WHAT?	"with A RULER"
21	chaturthi	dative	FOR WHOM?	"for THE SYSTEM"
22	panchami	ablative	FROM WHERE?	"from REST"
23	shashthi	genitive	WHOSE?	"THE CAR'S velocity"
24	saptami	locative	WHERE/WHEN?	"at THE SURFACE"

Kaala (Tense — WHEN)

#	Name	Tense	Signal	Example
25	bhuta-kaala	past	was, flew	"the ball FLEW"
26	vartamana-kaala	present	is, moves	"the car IS moving"
27	bhavishya-kaala	future	will	"it WILL reach"
28	vidhi-kaala	imperative	find, calculate	"FIND the energy"
29	kala	time (general)	—	temporal reference

Quantity / Counting

#	Name	English	Captures	Example
30	sankhya	number	numeric value	"1000 kg" → 1000
31	asprista-sankhya	untouched number	unbound number	floating "5"
32	matra	unit	measurement unit	kg, m/s, joule
33	rashi-bandha	quantity binding	number → concept	1000 → mass

Structure / Ordering

#	Name	English	Captures
34	krama	sequence	formula step chain
35	kramanusara	following order	dependency
36	poorva/purva	before	predecessor
37	paschat	after	successor
38	amsha	part	part-whole
39	adhikarana	container	locus
40	garbha	inner	nested
41	antya	end	terminal

Truth / Validity

#	Name	English	Captures
42	satya	truth	how established
43	mithya	false	rejected/negated
44	viraam	stop/period	grade boundary

Classification / Taxonomy

#	Name	English	Captures
45	vishesa	specialization	X is a kind of Y
46	lakshana	characteristic	X is a property of Y
47	karma	action/verb	node's action
48	vrnda	group	collection
49	avastha	state	condition
50	apeksha	dependency	depends on

Transformation / Philosophical

#	Name	English	Captures
51	janaka	generator	what creates this
52	prayoga	application	usage
53	rahita	without	absence ($yukta^{-1}$)
54	atikrama	crossing	exceeding limit
55	nirodha	cessation	stopping ($kriya^{-1}$)
56	paripurna	full	completeness
57	atita	beyond	transcended
58	bhanga	breaking	rupture
59	daatri	giver	source
60	vibheda	distinction	splitting ($abheda^{-1}$)
61	ahara	input	what feeds in

Dynamic = Composed from Static

Dynamic	Composition	Meaning
karma	$kriya \otimes kriya$	action of action
janaka	$janya^{-1}$	reverse of born-from
vibheda	$abheda^{-1}$	reverse of equivalence
rahita	$yukta^{-1}$	reverse of connection
nirodha	$kriya^{-1}$	reverse of action
mithya	$pratipaksha \otimes satya$	inverse of truth
satya	$siddha \otimes siddha \otimes \dots$	iterated proof

Part 8: Pratipaksha Grid

The pratipaksha floor — one concrete grid from the tensor.

Arithmetic Operations Grid (Floor 10)

addition	subtraction	mul	division	square	sqrt	half	double
addition	0	**1**	0	0	0	0	0
subtraction	**1**	0	0	0	0	0	0
multiplication	0	0	0	**1**	0	0	0
division	0	0	**1**	0	0	0	0
square	0	0	0	0	0	**1**	0
sqrt	0	0	0	0	**1**	0	0
half	0	0	0	0	0	0	0
double	0	0	0	0	0	0	**1**

Properties: symmetric (if $A \rightarrow B$ then $B \rightarrow A$), sparse, clean pairs.

Full Pratipaksha Pairs

Operation	Pratipaksha	Domain
addition $\leftrightarrow$ subtraction	add $\leftrightarrow$ sub	arithmetic
multiplication $\leftrightarrow$ division	mul $\leftrightarrow$ div	arithmetic
square $\leftrightarrow$ square-root	square $\leftrightarrow$ sqrt	power
half $\leftrightarrow$ double	half $\leftrightarrow$ double	scaling
logarithm $\leftrightarrow$ exponential	log $\leftrightarrow$ exp	transcendental
sine $\leftrightarrow$ arcsine	sin $\leftrightarrow$ arcsin	trigonometry
cosine $\leftrightarrow$ arccosine	cos $\leftrightarrow$ arccos	trigonometry
derivative $\leftrightarrow$ antiderivative	d/dx $\leftrightarrow$ $\int dx$	calculus
max $\leftrightarrow$ min	—	comparison
ceil $\leftrightarrow$ floor	—	rounding
vridhhi $\leftrightarrow$ kshaya	increase $\leftrightarrow$ decrease	signals
abheda $\leftrightarrow$ vibheda	sameness $\leftrightarrow$ difference	philosophy
yukta $\leftrightarrow$ rahita	connection $\leftrightarrow$ absence	philosophy
kriya $\leftrightarrow$ nirodha	action $\leftrightarrow$ cessation	philosophy
aarambham $\leftrightarrow$ antya	beginning $\leftrightarrow$ end	philosophy
phala $\leftrightarrow$ janya	output $\leftrightarrow$ input	structure

Part 9: Cross-Floor Walks

Within a floor: $\oplus$ = OR (accumulate 1s). Across floors: $\otimes$ = chain (output of one floor feeds as input to next).

The walker JUMPS to whichever floor it needs. Not sequential.

Walk: "inverse of what addition acts through"

Step	Floor	Row	Find 1s at
1	kriya (7)	addition	matra, arithmetic, vridhhi
2	pratipaksha (10)	vridhhi	kshaya

Result: kshaya (decrease). Two jumps, 60 floors untouched.

Walk: "solve $K = \frac{1}{2}mv^2$ for velocity"

Step	Floor	Action	Result
1	krama (34)	read formula	half, mul, square
2	pratipaksha (10)	invert each	double, div, sqrt
3	—	reverse order	sqrt, div, double
4	—	evaluate	$v = \sqrt{2K/m}$

Non-commutativity

Walk order	Path	Result
kriya $\otimes$ pratipaksha	addition to vridhhi to kshaya	inverse of mechanism
pratipaksha $\otimes$ kriya	addition to subtraction to ???	mechanism of inverse

Different answers. Floor order matters.

Janya + Phala Grids (formula inputs/outputs)

Janya floor (inputs):

mass	velocity	acceleration
KE-mantra	1	1
momentum-mantra	1	1
force-mantra	1	0

Phala floor (outputs):

kinetic-energy	momentum	force
KE-mantra	1	0
momentum-mantra	0	1
force-mantra	0	0

Both floors together: KE-mantra takes mass and velocity, produces kinetic-energy.

Part 10: The Pipeline

A question IS a tantra. Same structure as a mantra.

Mantra (formula)	Question
janya	mass, velocity
krama	half $\circ$ mul $\circ$ square
phala	250 (number)
temporary	variable bindings

Pipeline Stages

Stage	Input	Output	Mechanism	Type
construct	sentence	raw-graph	materialize, shabda-anveshana	transducer
assert	raw-graph	asserted-graph	assertion-bandha	—
refine	asserted-graph	refined-graph	fixpoint $\times 13$ sub-passes	endomorphism
expand	refined-graph	expanded-graph	PPR walk	morphism
detect	expanded-graph	intent-signals	signal detection	morphism
dispatch	intent-signals	answer	derive/count/viveka/anumana	—
emit	answer	proof	panchaavayava formatting	—

Each stage only ADDS 1s. Monotonicity: question graph only grows.

Discovery-97: Pipeline as Σ

Pipeline stages should have eval keys and pratipaksha:

Stage	Pratipaksha	Meaning
construct	generation	graph $\rightarrow$ sentence
refine	abstraction	richer $\rightarrow$ simpler
expand	compression	expanded $\rightarrow$ local
detect	synthesis	signals $\rightarrow$ graph
dispatch	question-generation	answer $\rightarrow$ question
emit	proof-checking	proof $\rightarrow$ answer

Gap: execution is in OCaml, not graph-native. Closing this makes questions fully invertible tantras.

Part 11: Traced Examples

Example 1: Physics — Kinetic Energy

Question: "mass is 5 and velocity is 10. find kinetic energy" Answer: "we find: kinetic-energy = 250"

Graded ring:

Grade	Sentence	Facts extracted
R_0	"mass is 5 and velocity is 10"	sankhya: mass=5, velocity=10
R_1	"find kinetic energy"	vidhi-kaala: KE=seek

Floor trace:

Stage	Floors read	Floors written	Effect
construct	(text)	sankhya, vidhi-kaala	mass=5, vel=10, KE=seek
assert	satya	satya	mass=established, vel=established
refine	sankhya, matra, yukta	rashi-bandha	mass=5, vel=10
expand	ALL (PPR)	janya, phala, krama	pulls in KE-mantra
detect	vidhi-kaala, rashi-bandha	intent	mode=derive, target=KE
dispatch	krama	sankhya, satya	evaluates formula
emit	all	(text)	panchaavayava proof

Krama evaluation:

Step	Operation	eval	Input	Output
1	square	square	velocity=10	100
2	multiplication	mul	mass=5, 100	500
3	half	half	500	250

Example 2: Count Addition (Graded Ring)

Question: "3 birds sat on a tree. 2 more came. how many birds are there in total" Answer: "5 bird there total"

Grade	Sentence	Signal	δ mapping
R_0	"3 birds sat on a tree"	sankhya=3	sat = sthita
R_1	"2 more came"	sankhya=2	more = vriddhi (increase)
R_2	"how many birds total"	prashna	total = sum

Count fold:

Step	acc	direction	value	result
start	0	—	shunya	0
R_0	$0 \oplus 3$	vriddhi = add	3	3
R_1	$3 \oplus 2$	vriddhi = add	2	5

Example 3: Count Subtraction (Pratipaksha)

Question: "10 birds sat on a tree. 3 flew away. how many birds are left" Answer: "7 bird left"

Grade	Sentence	Signal	δ mapping
R_0	"10 birds sat on a tree"	sankhya=10	—
R_1	"3 flew away"	sankhya=3	flew away = kshaya
R_2	"how many birds left"	prashna	—

Critical edge: $G[vriddhi][kshaya][pratipaksha] = 1$

Step	acc	direction	value	result
start	0	—	shunya	0
R_0	$0 \oplus 10$	vriddhi = add	10	10
R_1	$10 \ominus 3$	kshaya = sub	3	7

One edge on floor 10 determines add vs subtract.

Example 4: Comparison (Viveka)

Question: "ball-A has mass 5. ball-B has mass 3. which is heavier" Answer: "ball-A is viveka-max"

Grade	Sentence	Entity	Value
R_0	"ball-A has mass 5"	ball-A	mass=5
R_1	"ball-B has mass 3"	ball-B	mass=3
R_2	"which is heavier"	—	heavier = viveka-max

Cross-entity $\otimes$: select across grades, compare, return max.

Example 5: Force (Chain Derivation)

Question: "mass is 10 and acceleration is 5. find force" Answer: "we find: force = 50"

Step	Operation	Input	Output
1	multiplication	mass=10, accel=5	50

Mantra: force-mantra, krama = (multiplication mass acceleration), $F=ma$.

Part 12: Glossary

Ring Vocabulary

Symbol	Sanskrit	English	Role
$\oplus$	yukta	connection	additive, commutative
$\otimes$	kriya	action	multiplicative, non-commutative
0	shunya	emptiness	additive identity, fold seed
1	swarupa	self-nature	multiplicative identity
-1	pratipaksha	opposition	group inverse

Graph Vocabulary

Sanskrit	English	What it is
nigamana	conclusion	a node (truth-that-holds)
visheshanam	relation	edge type (one of 62 floors)
satya	truth	truth score
shabda	word	key-value metadata on nodes
tantra	program	declarative composition
mantra	formula	janya $\rightarrow$ phala via krama
krama	sequence	ordered operation chain
Σ	operation set	all nodes with eval keys

Panchaavayava (Five-Limbed Proof)

Step	Sanskrit	English	Role
1	pratijna	thesis	states givens
2	hetu	reason	states sought
3	udaharana	example	shows formula
4	upanaya	application	substitutes values
5	nigamana	conclusion	states answer